before:

$\dot{v} = M^{-1}(q)Bu$

$\bar{B} = M^{-1}(\phi^{-1}(0))\,B$

Approximation

$\dot{v} = \bar{B}u$

now:

$\dot{v} = M^{-1}(q)\,f(q,v,u)$ with $f(q,v,0) = 0$

linearize at $(q,v,u) = (0,0,0)$

$$\bar{B} = M^{-1}(\phi^{-1}(0))\left.\frac{\partial f(q,v,u)}{\partial \begin{pmatrix} q \\ v \\ u \end{pmatrix}}\right|_{q=v=u=0}$$

- $\bar{B}$ full rank
- $\frac{\partial f}{\partial q_i} = 0 \quad \forall i$ with $w_i > 1$

only q_i with $w_i = 1$ appear

Approximation

$$\dot{v} = \bar{B}\begin{pmatrix} q \\ v \\ u \end{pmatrix}$$

homogeneity $h(\Lambda_\lambda x, \Delta_\lambda u) = \lambda^{\tilde{r}}\Lambda_\lambda f(x,u)$

variable	homogeneity parameter
q	$r_i = w_i$
v	1
u	1

diag(1)

diag(1)

$\dot{v}_i = \bar{b}_i^T \Delta_\lambda u = \lambda \bar{b}_i^T u = \lambda^0 \Lambda_\lambda \bar{b}_i^T u$

$\bar{b}_i^T$: Reihe von $\bar{B}$

$\dot{v}_i = \bar{B}_i(\Lambda_\lambda q, \Delta_\lambda v, \Delta_\lambda u) = \lambda^{\tilde{r}}\Lambda_\lambda \bar{B}_i(q,v,u)$ ✓

X: $1, 1, 2, 3, \ldots$

diag(1) diag(1) diag(1)